

Report Preparation Guide

Basics

This document serves as an aide memoire to guide the production of the BRA-EUR report.

General design criteria

- the report is built using quarto and the R-ecosystem, the final output in form of a pdf version and accompanying html (quarto) book
- associated chapters are managed / included via `_quarto.yml`, supporting and releasable documents are provided (e.g. this aide-memoire)
- reserved folders
 - figures – keeps all embedded figures (e.g. maps), if required “tailored” figures may be composed using powerpoint (and save as): `__figures-handmade.pptx`
 - R – all R functions (once steady and industrialised)
 - data – the data to reproduce (and render) the report, ideally in (harmonised) PBWG format
 - docs – reserved, this folder is used for the hosting of the html report on github
 - (gitignored) `_book-local` – to support local rendering of contributors

This report is produced by various individuals with different programming backgrounds. The following coding style shall serve to increase the readability of code and its review/troubleshooting.

Some of the work may be outsourced to another project aiming at producing a supporting R-package.

Coding principles

- the principle paradigm is i.) read (analytic or performance-related) data, ii.) process and prepare the data for the analysis at hand, and iii.) “communicate” (either as table or visualisation) the results
- these steps shall be performed starting from the agreed harmonised “PBWG” for-

- mat or a proposed version (if trialled first in BRA-EUR)
- associated functions shall be **generic**, i.e., not building on specifics of the regional data
- specific folders – c.f. above
- the report and supporting repo will be “public” - use .gitignore
- gitignore “large” source data/files to avoid issues with github, share the files via joint folders/file transfer
- coding style
 - variables and function-names: lower-case with underscore, e.g. `asmas`, `calc_travel_time`
 - try to name what it is or does, e.g. taxi-in times: `txits`, ggplot function: `plot_fancy_thing`
 - dataframe/tibble: column-names upper-case for all standard aviation terms, e.g. actual taxi-in time: `TXIT`, use underscore prefix/postfix, e.g. `TOT_TXIT`
 - quarto tags: use hyphen, e.g. `fig-cool-plot`, `sec-herehappensomething`
 - file names: use hyphen

Overall Data Flow/Process

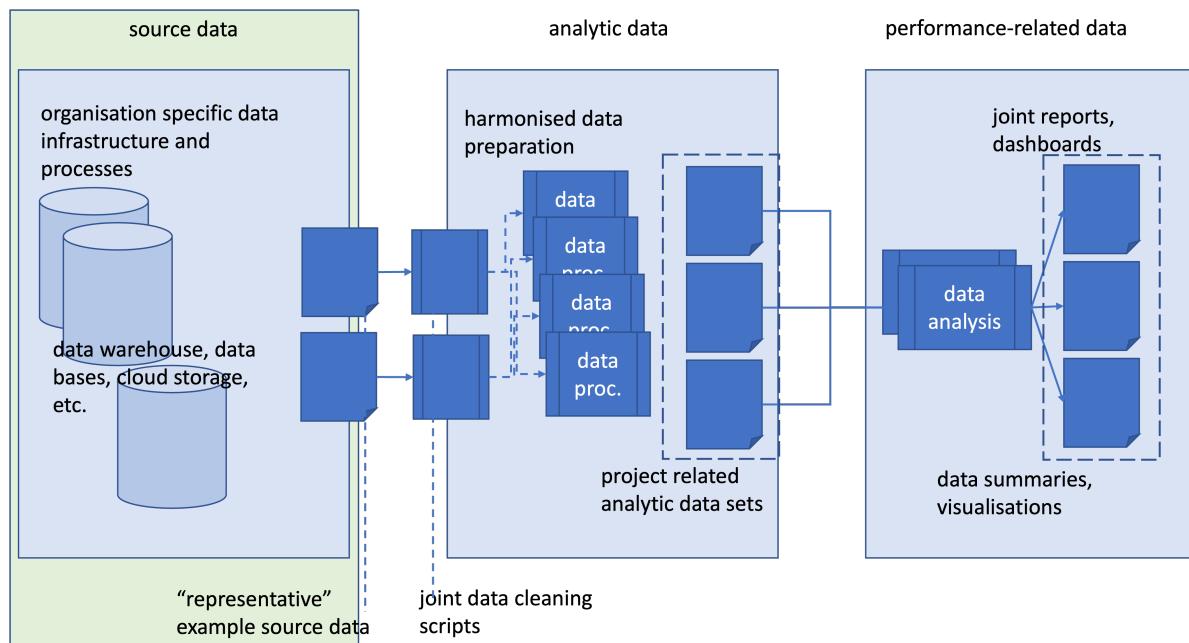
Both groups process and expose their data in different systems and technologies.

While an open exchange of the underlying **source data** is a means to continue with the harmonisation of the data processes, the goal is to arrive at a set of harmonised (PBWG) data sets for the common indicators.

Accordingly, the source data extraction may differ between the parties, e.g. Brazil/DECEA is developing endpoints for API queries, ECTRL/PRU is limited by the on-going cloud transition and currently access their data via a classical data warehouse/SQL.

The BRA-EUR report builds on the PBWG work and will propose changes to the latter during further developments.

```
knitr::include_graphics("./figures/data-source-analytic-perfrelated.png")
```



The report broadly distinguishes between network-level and airport-level reporting.

Network-level Data

Airport-level Data

APDF - Airport Data Flow

The APDF format represents a compact airport-level format from which the common GANP indicators can be calculated, including variants.

The ADPF data sets ensure also a decent basis for the on-going developments.

TODO - describe format